

Predict Electricity Usage Using PG&E Load Data

Prepared for Data Science
& Big Data Final Project

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AGENDA

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01

Introduction

Why electricity load forecasting?

In recent years, California's energy market has become increasingly volatile due to the growing impacts of drought, wildfires, and rapid population growth. These factors have placed immense pressure on utilities like PG&E to maintain a stable and reliable energy supply amid rising and fluctuating demand.

Thus, predicting electricity demand accurately both temporally and geographically becomes an important task to regional utilities.



2019, after having enough power outages, Kaia and her college roommates dressed up as PG&E and went to every dorm room to turn off their light

Traditional Energy Load Forecasting vs ML

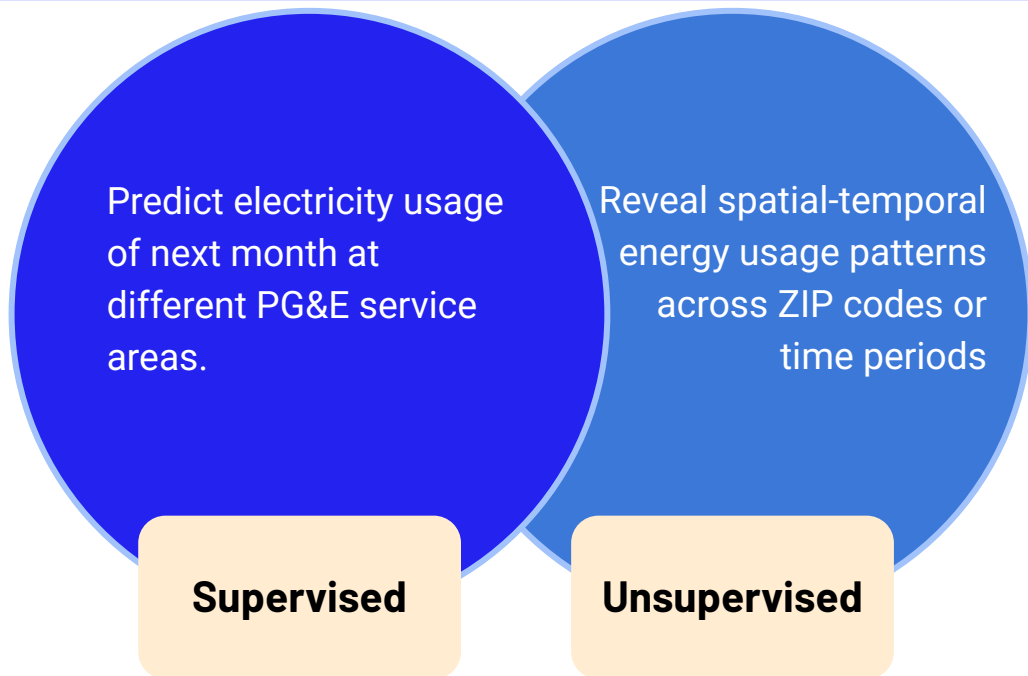
- Most traditional models for predicting energy usage often rely on time series analysis techniques and regression models. They do not need a lot of historical data and can construct the explicit relationship between energy consumption and its influencing factors, such as lagged observation, temperature, GDP, and population (Deb et al., 2017).
- In contrast with conventional models, AI-based forecasting models do not rely on the explicit relationship between energy consumption and its influencing factors but learn from a large amount of historical data for prediction instead.



Which methodology is better for predicting monthly energy load trend?

How do we combine the best of two worlds to perform forecasting and uncover interesting utilization patterns?

Goals



Our Data

PG&E Energy Data Request-Public Data Sets

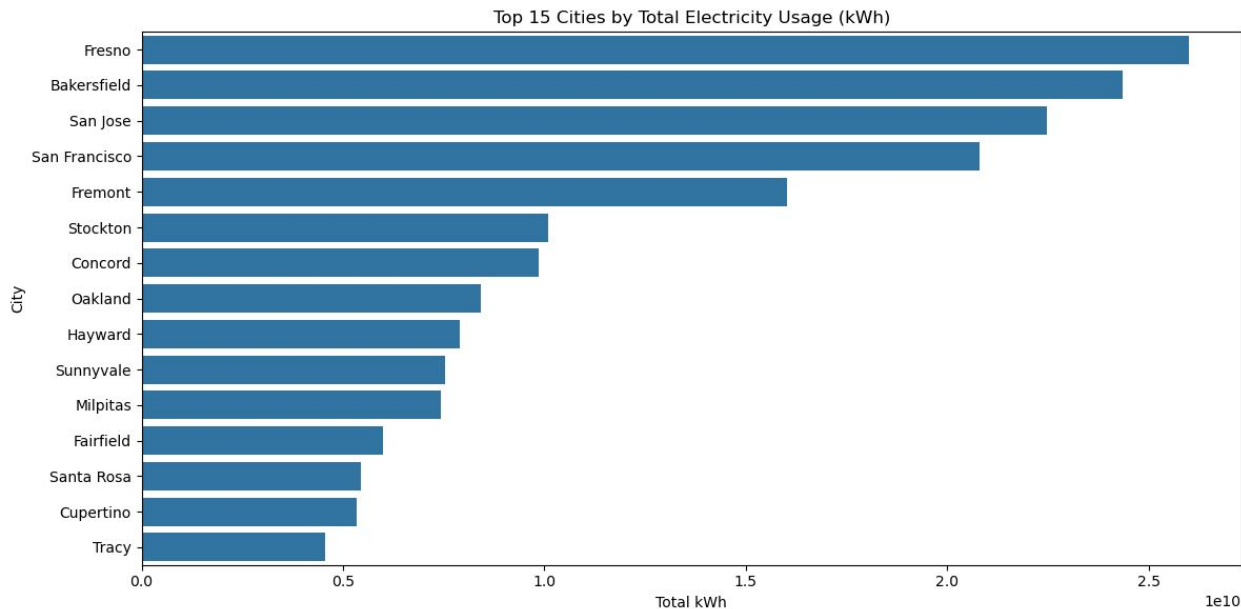
- **Fields:** Zip Code, Month, Year, Customer Class, Total Customers, Total KWH, Average KWH
- **Time Range:** From 2020 to 2024, monthly
- **Length:** 9000~10,000 per year, 45,000 rows in total

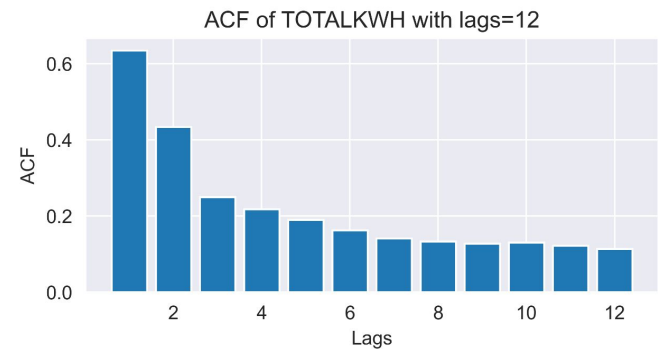
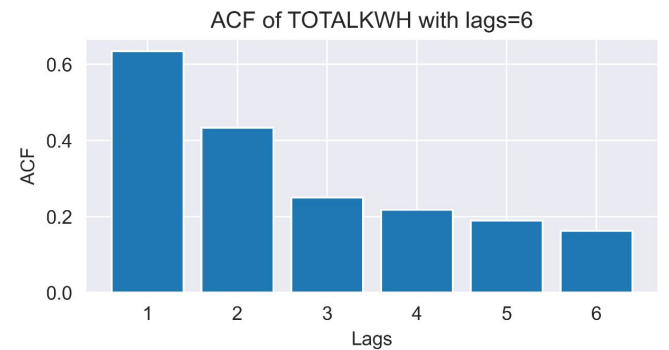
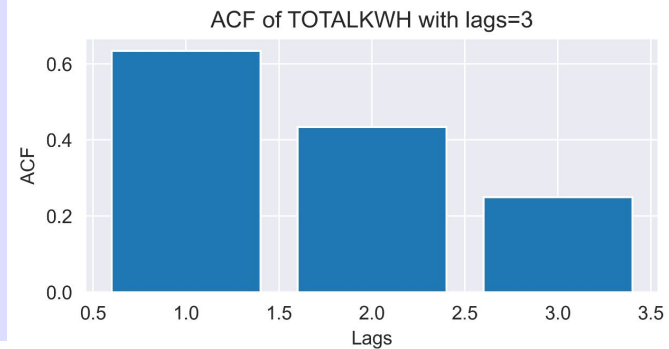
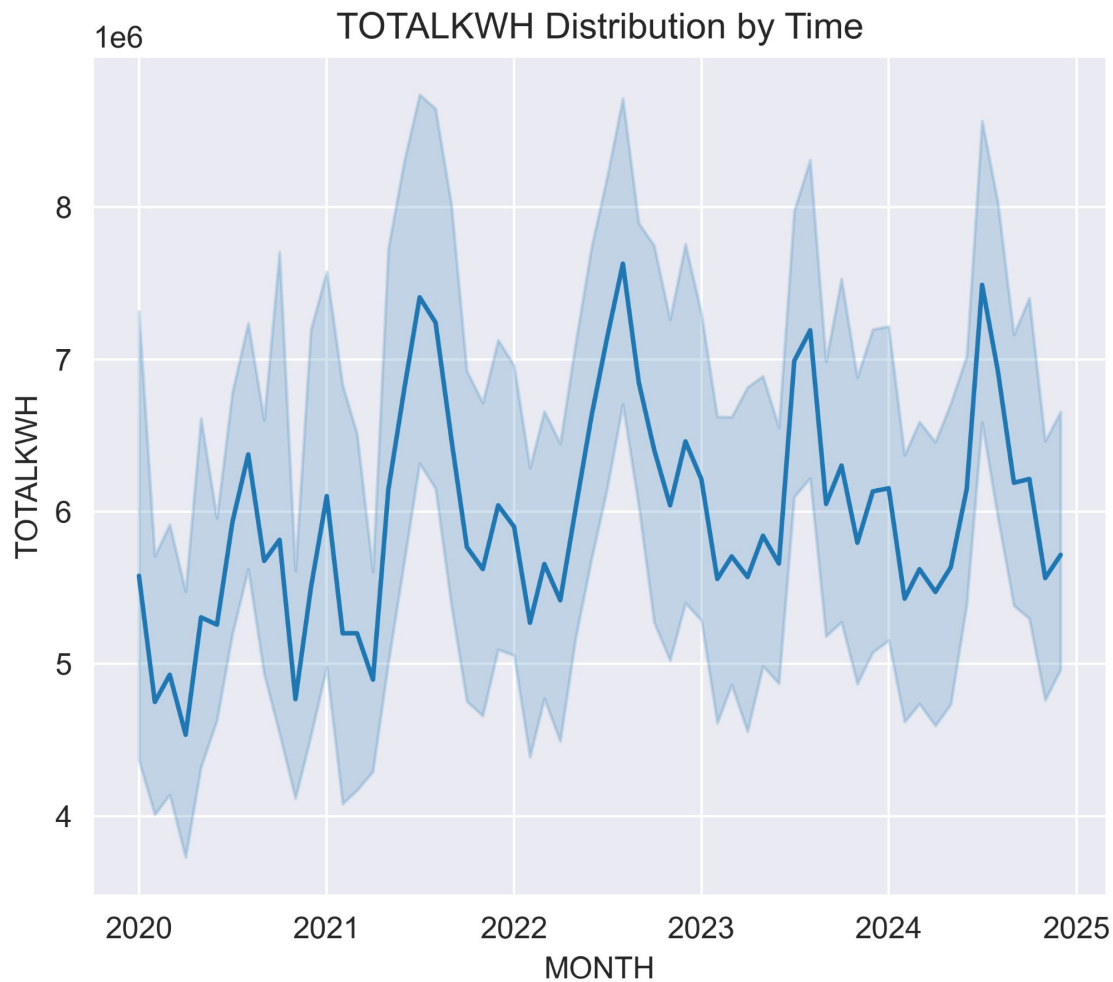
Weather Data Scraping from Weather Underground

- **Fields:** Weather Station, Temperature, Dew Point, Precipitation, Wind, Sea Level Pressure
- **Time Range:** From 2020 to 2024, monthly
- **Length:** About 150 weather stations. About 9,000 rows in total

EDA

Few cities have significantly higher electricity consumption compared to the others



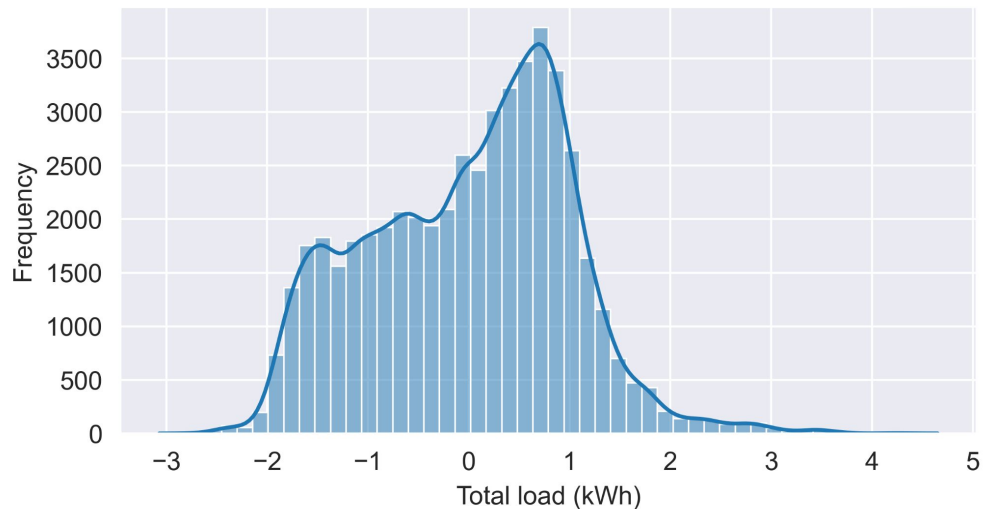
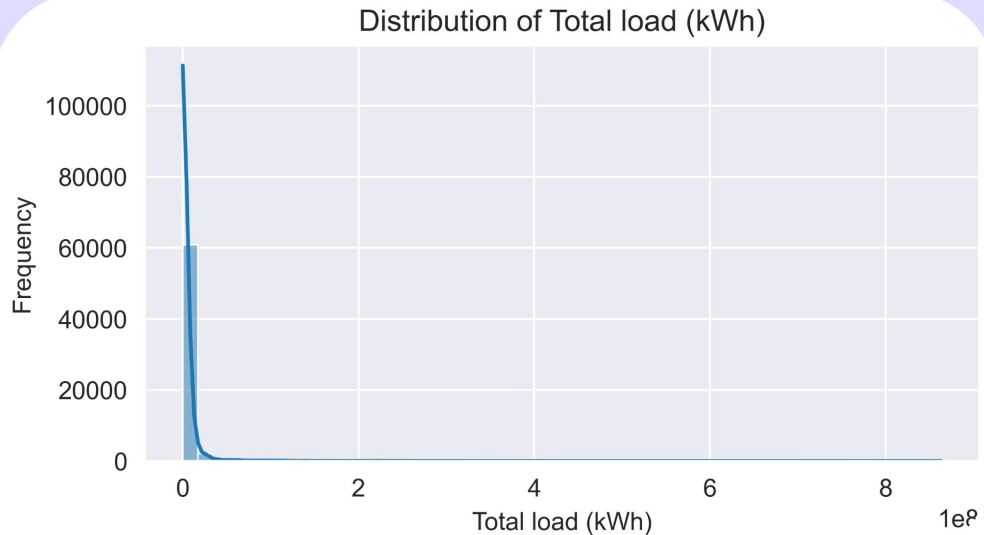


Issue with Data

Uneven distribution in our target:
Strong Right Drag

Our action:

- Apply Yeo-Johnson transformation with standardization on highly-skewed features
- Apply standardization on other features



02

Methodology

Data Preprocessing

Feature Selection:

- Drop features with little correlation (<0.1) with target
- Deal with high multi-linearity between temperature and dewpoint: drop dewpoint, according to domain knowledge
- Calculate monthly cyclic parameters: month_sin and month_cos

Data Splitting:

- Train/Test: 8/2
- Cross Validation: Time Splitting with Fold=5

	Model Name	Model Strength	Optimize: Hyperparameter Tuning
Supervised ML Models	Linear Regression	Simplicity & interpretability	X
	Elastic Net	Regularization for LR	alpha, l1_ratio
	Random Forest Regressor	Tree-based & Interpretability	max_depth, max_features, min_samples_split, min_samples_leaf
	XGBoost Regressor	Accuracy	booster, n_estimator, learning_rate, gamma
	Long Short-Term Memory (LSTM)	Long-Term Dependencies	hidden_size, num_layers, early stopping, model structure
Unsupervis ed ML Models	K-Means Clustering	Interpretability & efficiency	Number of clusters
	Time-Series Shape-Based Clustering	Focus on Pattern Similarity	Number of clusters

Evaluation Metrics

Model Evaluation:

- R-Square: Measure the explanatory power of the model
- MSE: Evaluate overall model accuracy
- RMSE: Report error in original units

Business Report:

- MAE: Monthly predictions differ from actual values by MAE kWh

03

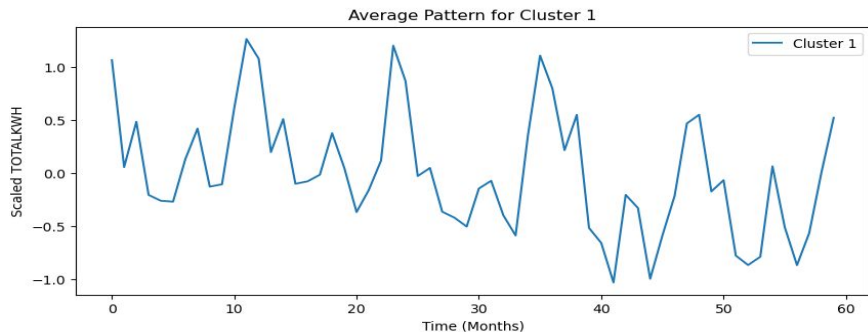
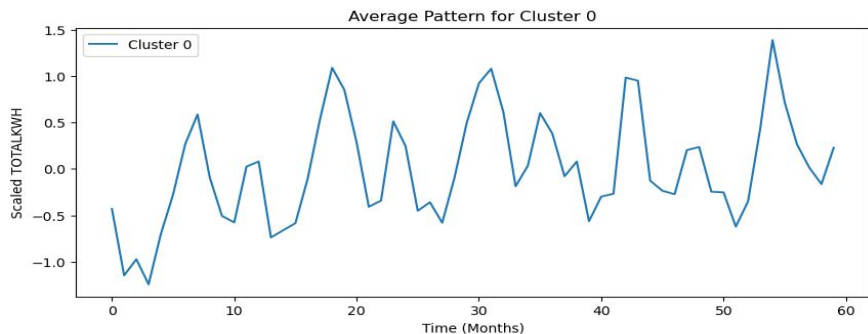
Results

Model Name	Cross Validation	Testing Set (Inverse Transformed)
Linear Regression	R-Square: 0.72 ± 0.01 MSE: 0.28 ± 0.01 RMSE: 0.53 ± 0.01	R-Square: 0.14 MAE: 3,536,064.14
Elastic Net	R-Square: -0.0004 ± 0.0002 MSE: 0.29 ± 0.02 RMSE: 0.54 ± 0.02	R-Square: 0.03 MAE: 4,048,387.82
Random Forest Regressor	R-Square: 0.87 ± 0.02 MSE: 0.13 ± 0.02 RMSE: 0.35 ± 0.03	R-Square: 0.56 MAE: 2,203,318.20
XGBoost Regressor	R-Square: 0.85 ± 0.03 MSE: 0.14 ± 0.02 RMSE: 0.37 ± 0.03	R-Square: 0.49 MAE: 2,318,009.23
Long Short-Term Memory (LSTM)	R-Square: 0.45 ± 0.15 MSE: 0.63 ± 0.06 RMSE: 0.79 ± 0.04	R-Square: 0.22 MAE: 3,606,711.32

Model Performance Observation

- Linear Regression, Elastic Net: CV $R^2 \approx 0.72$, ~ 0 ; test $R^2 \approx 0.14$, 0.03 ; high errors.
- Tree-based ensemble models best capture zipcode-level monthly electric load patterns.
- Random Forest, XGBoost: CV $R^2 \approx 0.87$, 0.85 ; test $R^2 \approx 0.56$, 0.49 ; lowest errors. Random Forest is more recommended, considering similar performance and lower cost.
- The monthly data has weak temporal autocorrelation. Therefore, LSTM model is poor in this task.

Unsupervised Learning



K Means

K Tested: [2, 3, 4, 5, 6]
Best K: 2
Silhouette Score: 0.1646

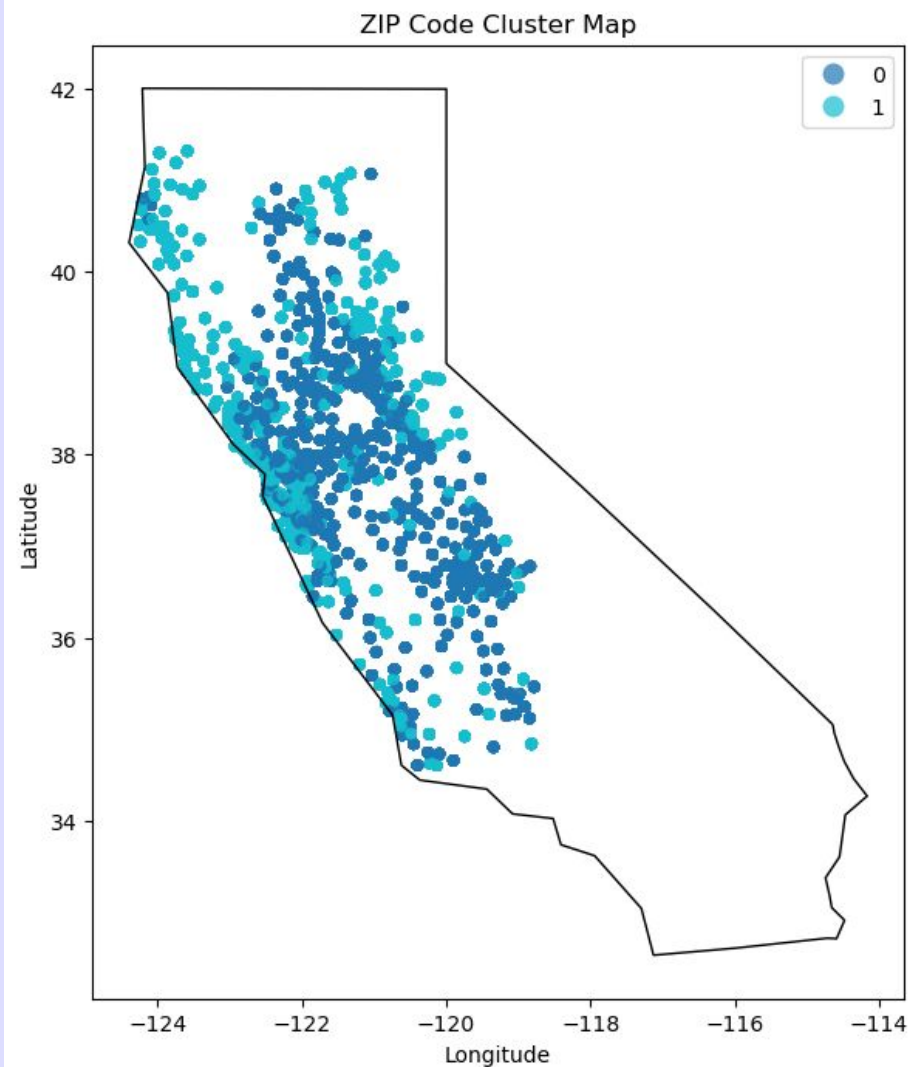
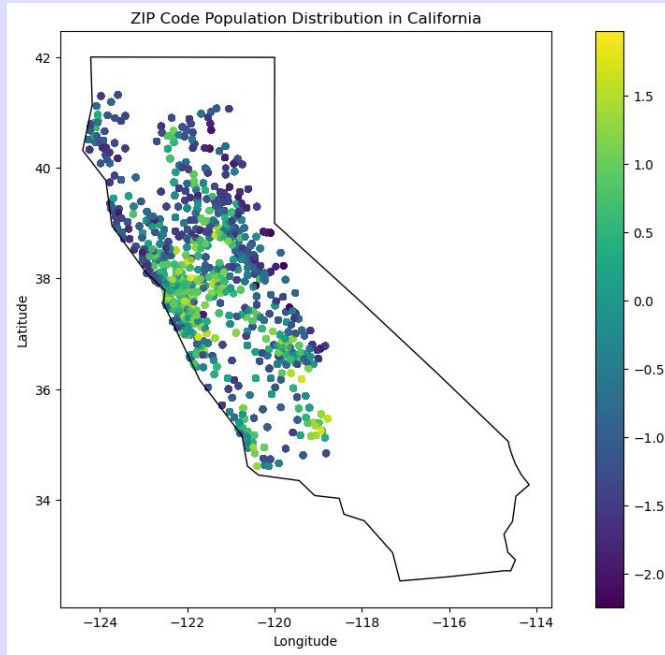
**DTW & Soft
DTW**

K Tested: [2, 3, 4]
Best K: 2 (with soft dtw)
Silhouette Score: 0.2235

Cities in Cluster 0 have electricity load peak in summers, with smaller peaks in winters

Cities in Cluster 1 have major peaks in winters, and no other significant peaks

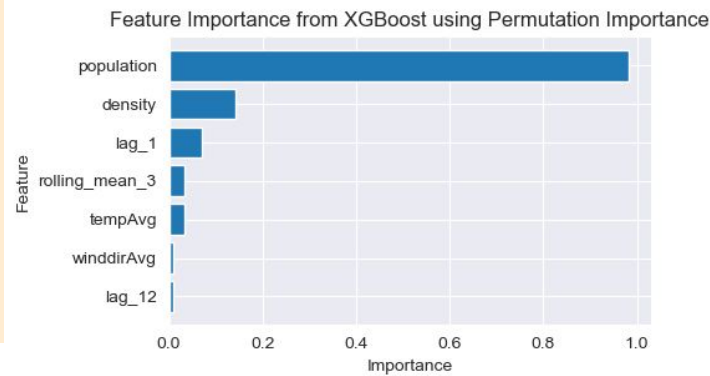
The cluster map overlaps significantly with the population distribution, indicating that the overall trends of energy consumption across the region is mostly determined by its population volume



04

Discussion / Conclusion

Major Findings



- Population is the most influential predictor of electricity consumption as well as time pattern of utilization (more populated→summer peaks; less populated→winter peaks).
- Time series historical data may show as a weaker predictor because two groups of cities have complete opposite seasonal energy usage patterns.
- There is a strong short-term persistence in the series, which could be valuable for local short-term projection.

Future Steps

- Investigate the impact of incorporating electricity usage types (e.g., residential, commercial, industrial) to improve model precision.
- Explore obtaining daily consumption data to enable fine-grained day-ahead or rolling daily forecasts.
- Assess how daily prediction models compare to monthly models in capturing short-term variability and improving responsiveness.
- Split the dataset using clustering and build separate prediction models for each cluster.

THANK YOU

Kaia & Serena